Cosmological Critical Coherence Index

A Structural-Stress Observable for Cosmic Evolution

Christof Krieg
MAAT Project – Independent Research, Germany
maat-research.com

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Abstract

We introduce the Cosmological Critical Coherence Index ($\mathrm{CCI}_{\mathrm{cos}}$), a dimensionless diagnostic designed to quantify the balance between expansion-driven stress and coherence-preserving structure growth in cosmic evolution. The index combines the normalised expansion rate $E(z) = H(z)/H_0$, a redshift-dependent activity factor, a linear-growth coherence proxy $D(z)$, and a structural imbalance term.

Using Planck-2018 ΛCDM parameters and a Cosmic Chronometer $H(z)$ compilation, we compute a first observational projection of $\mathrm{CCI}_{\mathrm{cos}}(z)$. In the reference model, the normalised index rises from 1 at $z = 0$ to approximately 6.66 at $z \simeq 1$, 22.6 at $z \simeq 2$, and 8.04×10^2 at $z = 10$. The chronometer projection follows the same increasing trend within the current data scatter.

The result should be interpreted as an observable proposal rather than a new cosmological model: $\mathrm{CCI}_{\mathrm{cos}}$ does not replace $H(z)$, $D(z)$, or standard parameter inference. Instead, it packages their relative imbalance into a single structural-stress history.

1 Status and Scope

This paper defines and evaluates a diagnostic observable. It does not introduce new field equations, fit new cosmological parameters, or claim evidence against ΛCDM. The purpose is narrower:

> *Can the expansion history and structure-growth history be combined into a dimensionless structural-stress index?*

The construction follows the general Critical Coherence Index (CCI) logic used in the structural-selection series: destabilising activity is placed in the numerator, while coherence-preserving structure appears in the denominator. In cosmology, the proposed diagnostic is not a replacement for precision likelihood analyses. It is a compressed observable that may be useful for visualising when cosmic evolution is expansion-dominated, growth-dominated, or structurally imbalanced.

> *While $H(z)$ describes the expansion history, $\mathrm{CCI}_{\mathrm{cos}}(z)$ provides a complementary structural-stress history.*

2 Definition

We define the Cosmological Critical Coherence Index by

$$\boxed{\mathrm{CCI}_{\mathrm{cos}}(z) = \frac{\Gamma_{\mathrm{inst}}(z)\,\Gamma_{\mathrm{prod}}(z)\;\left[1 + \kappa U_{\mathrm{struct}}(z)\right]}{1 + \Gamma_{\mathrm{coh}}(z) + \epsilon}} \tag{1}$$

with

$$\Gamma_{\rm inst}(z) = E(z) = \frac{H(z)}{H_0}, \qquad \Gamma_{\rm prod}(z) = 1 + z. \tag{2}$$

The coherence sector is represented by the linear growth factor and the normalised fluctuation amplitude,

$$\Gamma_{\rm coh}(z) = \frac{1}{2}\left[D(z) + \frac{\sigma_8(z)}{\sigma_{8,0}}\right], \qquad \sigma_8(z) = \sigma_{8,0} D(z). \tag{3}$$

Thus, to leading order in the present proxy implementation,

$$\Gamma_{\rm coh}(z) = D(z), \tag{4}$$

but Eq. (3) keeps the connection to observable structure-growth quantities explicit.

The structural imbalance term is

$$U_{\rm struct}(z) = \frac{|E(z) - \Gamma_{\rm coh}(z)|}{E(z) + \Gamma_{\rm coh}(z) + \epsilon}. \tag{5}$$

Unless stated otherwise, we use

$$\kappa = 1, \qquad \epsilon = 10^{-9}.$$

The present diagnostic contains no fitted MAAT sector weights. Equivalently, the reduced CCI projection uses a unit structural weighting convention,

$$\lambda_{\rm inst} = \lambda_{\rm prod} = \lambda_{\rm coh} = \lambda_U = 1,$$

with $\kappa = 1$ setting the relative strength of the imbalance correction. Thus the numbers reported below are not the result of a parameter fit. Sensitivity to non-unit κ and to calibrated sector weights is left for future multi-probe work. For presentation, we also use the normalised index

$$\widehat{\rm CCI}_{\rm cos}(z) = \frac{{\rm CCI}_{\rm cos}(z)}{{\rm CCI}_{\rm cos}(0)}. \tag{6}$$

3 Operational Conventions

Three conventions are fixed in this first implementation.

Weights. The scalar diagnostic in Eq. (1) uses unit weights and one imbalance coupling $\kappa = 1$. This is intentionally minimal: it prevents the observable from becoming a fitted score and makes the resulting curve a direct projection of $H(z)$ and $D(z)$. A later full MAAT-cosmology functional may introduce calibrated weights λ_a for multiple sectors, but those weights are not inferred in the present paper.

Cutoffs and regimes. The shaded redshift bands in Fig. 1 are interpretive guides only. They are not fitted cutoffs and are not used in the calculation of ${\rm CCI}_{\rm cos}$. In this note the only numerical cutoff is the regularisation $\epsilon = 10^{-9}$, used to avoid division by zero. Future work may define data-driven transition redshifts by extrema or curvature changes of $\mathrm{d}\log \widehat{\rm CCI}_{\rm cos}/\mathrm{d}z$, or by joint fits to growth and distance probes.

Future V and R measurements. The present ${\rm CCI}_{\rm cos}$ compresses expansion stress and linear growth coherence only. It does not yet measure the full MAAT connectivity and robustness sectors. Operationally, a cosmological connectivity sector could be defined from correlation observables such as

$$V_{\rm corr}(z) \sim \mathcal{N}_V[\xi(r,z),\ P(k,z),\ C_\ell,\ f\sigma_8(z)]\,, \tag{7}$$

where $\mathcal{N}_V$ denotes a dimensionless normalisation relative to a chosen reference model. A robustness sector could be defined from consistency and stability margins, for example

$$R_{\text{robust}}(z) \sim \left[1 + \chi_H^2(z) + \chi_{\text{growth}}^2(z) + \Pi_{\text{path}}(z)\right]^{-1}, \tag{8}$$

where Π_{path} penalises pathological regimes such as negative effective density, ghost-like stability violations, or large departures from joint distance–growth consistency. These definitions are stated as operational targets for the next-stage multi-probe analysis, not as quantities already measured in the present chronometer-only projection.

4 Cosmological Implementation

The reference background is a flat Planck-2018 ΛCDM cosmology [1] with

$$H_0 = 67.4\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}, \qquad \Omega_m = 0.315, \qquad \Omega_\Lambda = 0.685, \qquad \sigma_{8,0} = 0.811. \tag{9}$$

The expansion function is

$$E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}. \tag{10}$$

The growth factor is approximated by the standard Carroll–Press–Turner growth suppression formula [9]:

$$D(z) = \frac{g(z)}{g(0)(1+z)}, \tag{11}$$

where

$$g(z) = \frac{5\Omega_m(z)}{2}\left[\Omega_m(z)^{4/7} - \Omega_\Lambda(z) + \left(1 + \frac{\Omega_m(z)}{2}\right)\left(1 + \frac{\Omega_\Lambda(z)}{70}\right)\right]^{-1}. \tag{12}$$

For the observational projection we use a Cosmic Chronometer $H(z)$ compilation based on the standard chronometer literature [2, 3, 4, 5, 6, 7, 8]. Since the chronometer data provide $H(z)$ but not direct matched measurements of $D(z)$ at the same redshifts, the observed projection uses

$$E_{\text{obs}}(z) = H_{\text{obs}}(z)/H_0$$

in the instability sector while keeping the ΛCDM growth proxy $D_{\Lambda\text{CDM}}(z)$ in the coherence sector. This makes the comparison a first diagnostic projection, not a full joint growth-expansion fit.

5 Results

5.1 Reference Projection

The ΛCDM reference projection is shown in Fig. 1. The diagnostic rises rapidly toward high redshift because the expansion/activity sector grows while the linear-growth coherence proxy decreases.

Table 1: Representative values of the normalised cosmological CCI in the Planck-2018 ΛCDM reference projection.

z	$E(z)$	$D(z)$	$\widehat{\text{CCI}}_{\text{cos}}(z)$
0.0	1.000	1.000	1.000
1.0	1.791	0.607	6.663
2.0	3.035	0.417	22.604
5.0	8.279	0.212	159.755
10.0	20.493	0.115	803.804

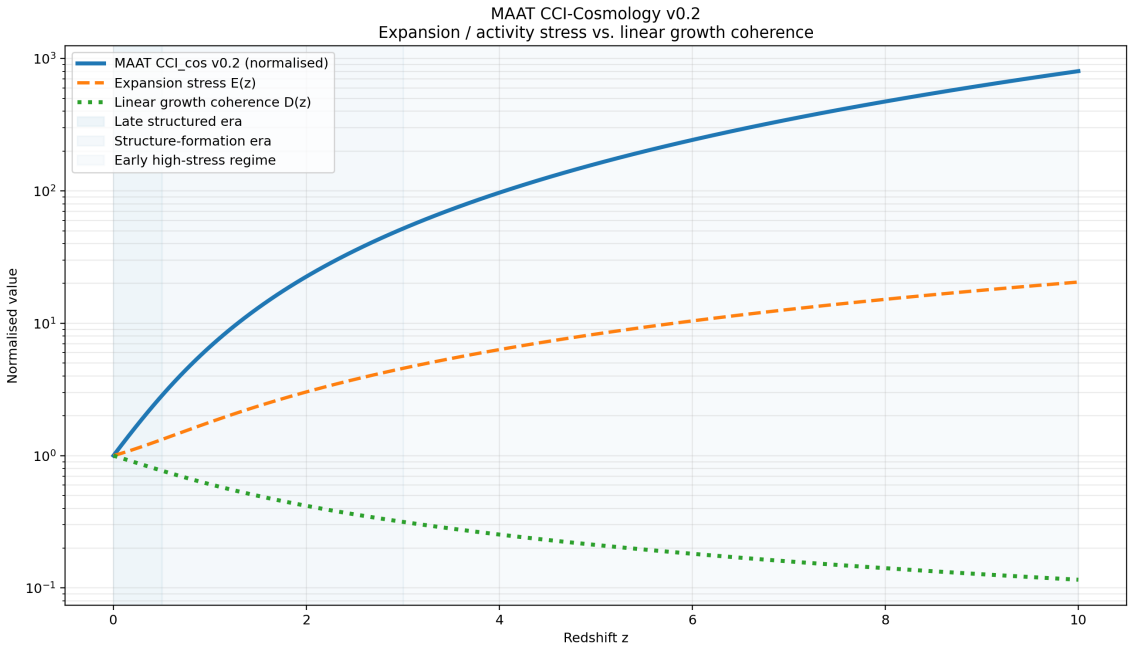

Figure 1: Planck-ΛCDM projection of the normalised cosmological CCI. The index compares expansion/activity stress with the declining linear growth coherence proxy. The shaded regions are interpretive guides, not fitted phase boundaries.

5.2 Cosmic Chronometer Projection

Figure 2 compares the Cosmic Chronometer measurements with the Planck-ΛCDM reference curve. The dataset used here contains 31 chronometer points over $0.07 \leq z \leq 1.965$.

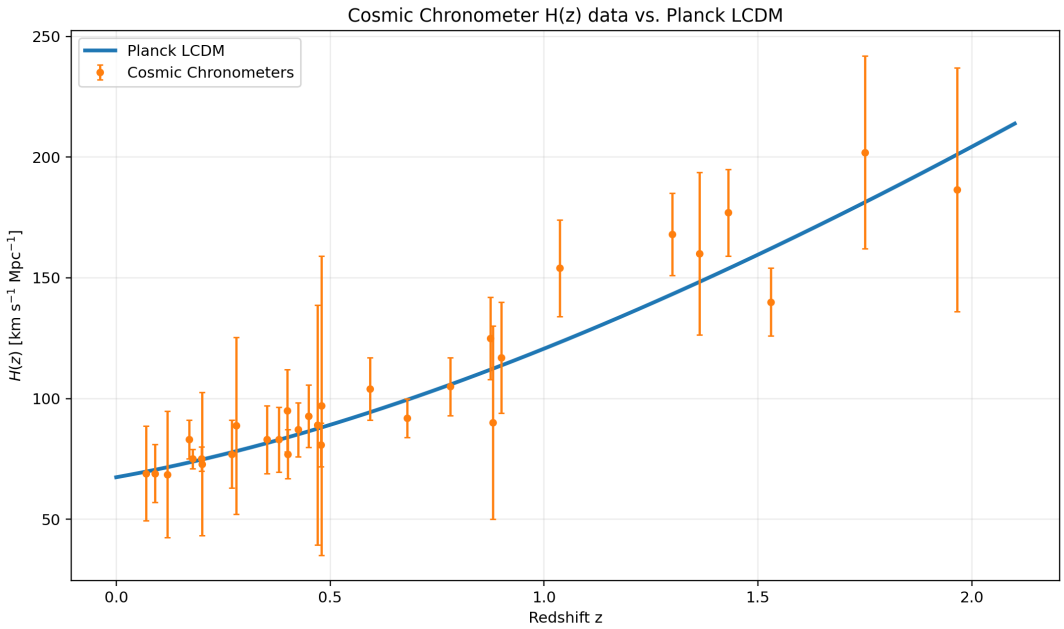

Figure 2: Cosmic Chronometer $H(z)$ measurements compared with the Planck-ΛCDM reference expansion history.

The corresponding CCI projection is shown in Fig. 3. The observationally projected index

increases from approximately 1.15 at $z = 0.07$ to approximately 20.0 at $z = 1.965$ in the normalised convention of Eq. (6). This trend is consistent with the interpretation that earlier epochs are structurally more expansion-stressed relative to late-time growth coherence.

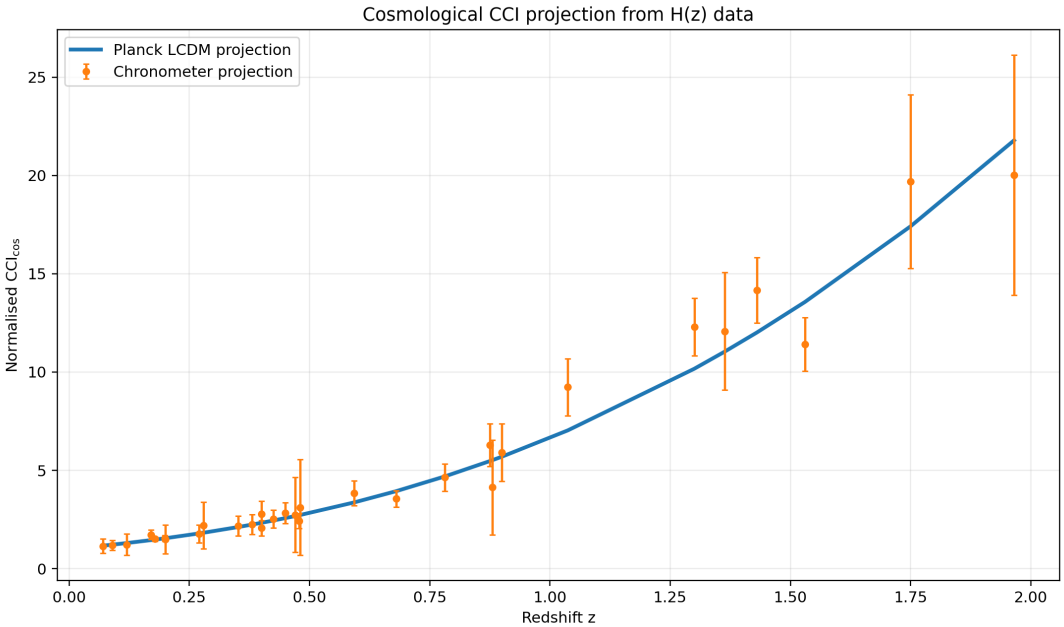

Figure 3: Observed projection of $\widehat{\mathrm{CCI}}_{\cos}(z)$ from Cosmic Chronometer $H(z)$ data compared with the Planck-ΛCDM reference. The growth-sector input is kept at the ΛCDM proxy level, so this is an expansion-driven diagnostic projection.

5.3 Residuals

The residuals relative to the ΛCDM projection are shown in Fig. 4. In the present dataset, the mean normalised CCI residual is approximately 0.22 and the root-mean-square residual is approximately 1.01. The largest absolute $H(z)$ pull in the chronometer sample is about 1.58σ. No statistically significant deviation from the reference model is observed at the current precision level.

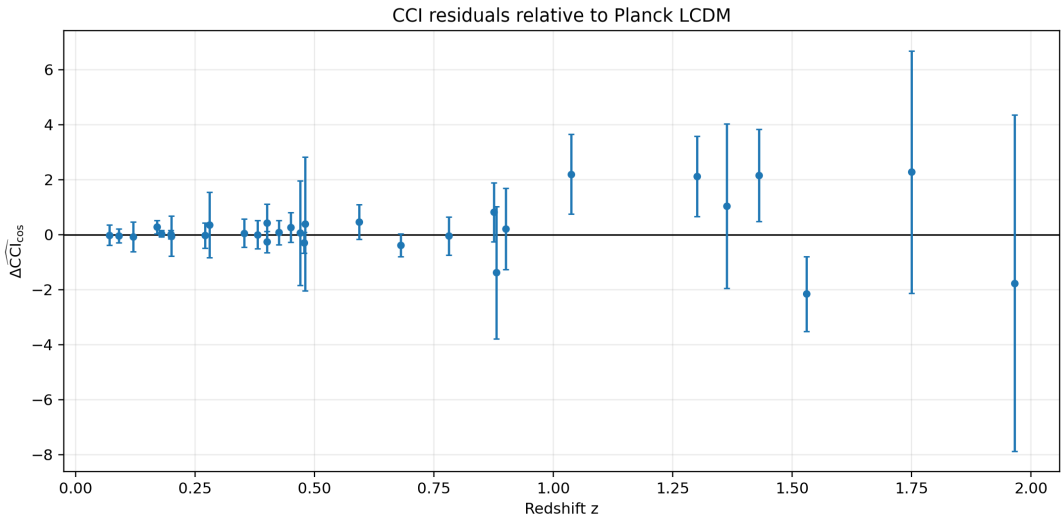

Figure 4: Residuals of the observed CCI projection relative to the Planck-ΛCDM reference. The current scatter is dominated by chronometer $H(z)$ uncertainties and should not be interpreted as evidence for a new cosmological component.

6 Interpretation

The diagnostic separates two roles that are usually discussed through different observables:

- $H(z)$ measures the expansion history,
- $D(z)$ and $\sigma_8(z)$ track the growth and coherence of structure,
- $\mathrm{CCI}_{\cos}(z)$ measures their structural imbalance.

In this language, the early universe is not merely "larger $H(z)$". It is a regime in which expansion/activity stress dominates over the coherence proxy available to linear structure growth. Late-time cosmic evolution then appears as a transition toward a lower-stress, coherence-supported regime.

The central interpretation is therefore:

> *Expansion defines the dynamical background; structure growth defines the coherence response;* $\mathrm{CCI}_{\cos}$ *measures their imbalance.*

Equivalently, while $H(z)$ describes the expansion history, $\mathrm{CCI}_{\cos}(z)$ provides a complementary structural-stress history.

7 Relation to the MAAT Framework

The cosmological CCI is a reduced observable, not the full MAAT structural functional. In MAAT language, the numerator of Eq. (1) captures expansion-driven activity and stress, while the denominator captures coherence-preserving structure growth. The missing sectors are equally important: constraint consistency, robustness, anisotropy, nonlinear clustering, and observational systematics are not yet included.

A fuller MAAT-cosmology functional would replace the present scalar projection with a multi-sector object such as

$$\mathcal{F}_{\cos} = F(H_{\mathrm{res}}, B_{\mathrm{cons}}, S_{\mathrm{act}}, V_{\mathrm{corr}}, R_{\mathrm{robust}}, \Sigma_{\mathrm{geom}}), \tag{13}$$

where the present $\mathrm{CCI}_{\cos}$ would act as one compressed stress diagnostic.

8 Limitations

The present implementation has several limitations:

- the chronometer projection uses $H(z)$ data only;
- growth information is supplied by an approximate ΛCDM proxy;
- unit weights and $\kappa = 1$ are fixed by convention rather than fitted;
- no data-driven regime cutoffs are inferred;
- the full MAAT connectivity (V) and robustness (R) sectors are specified only as operational targets in Sec. 3, not measured in the present chronometer-only analysis;
- BAO, supernovae, redshift-space distortions, weak lensing, and DESI data are not included;
- the growth factor is approximate and should be replaced by a Boltzmann-code or growth-ODE calculation in precision work;
- no parameter inference, covariance modelling, or model comparison is performed;
- the regime shading in Fig. 1 is interpretive, not a statistically fitted transition structure.

9 Reproducibility

The code and generated artifacts for this diagnostic are archived in the public repository:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/cosmological_cci

Within the repository layout, the analysis can be reproduced with:

```
python3 maat_cci_cosmology_v02.py
```

This command regenerates the model grid, the chronometer comparison table, and all four figures used in the paper. The generated figures are written to the `plots/` subdirectory.

The local files used for this note are:

- `maat_cci_cosmology_v02.py`,
- `maat_cci_cosmology_v02_chronometers.csv`,
- `maat_cci_cosmology_v02.csv`,
- `maat_cci_cosmology_v02_data_comparison.csv`,
- `plots/maat_cci_cosmology_v02_plot.png`,
- `plots/maat_cci_cosmology_v02_Hz_comparison.png`,
- `plots/maat_cci_cosmology_v02_data_comparison.png`,
- `plots/maat_cci_cosmology_v02_residuals.png`.

These files define a closed reproducibility loop:

$$\text{Planck parameters} + \text{Chronometer } H(z) \;\rightarrow\; \mathrm{CCI}_{\cos}(z) \;\rightarrow\; \text{figures and residuals.}$$

10 Data Provenance and Reuse

The Planck parameter values are quoted from the Planck-2018 cosmological parameter analysis [1]. The Cosmic Chronometer values are a small literature-derived $H(z)$ table compiled from the cited chronometer sources. The present work does not claim ownership of the original observational measurements, and reuse of those measurements remains subject to the terms of the original publications, journals, and collaborations.

All CSV files and figures generated here are derived analysis artifacts of this project. They are provided only to make the diagnostic calculation transparent and reproducible. No endorsement by the Planck Collaboration or by the authors of the chronometer measurements is implied.

11 Outlook

Natural next steps are:

- include BAO, DESI, supernova, redshift-space-distortion, and weak-lensing data;
- replace the approximate growth factor with a numerical growth equation or Boltzmann-code pipeline;
- test whether $\mathrm{CCI}_{\cos}$ reacts systematically to S_8 and H_0 tensions;
- compare ΛCDM, $w_0 w_a$CDM, early-dark-energy, and modified-gravity models in the same structural-stress representation;
- extend the scalar CCI diagnostic into the full multi-sector MAAT cosmology functional.

12 Conclusion

We introduced the Cosmological Critical Coherence Index as a dimensionless structural-stress observable for cosmic evolution. In a Planck-ΛCDM reference projection, the index grows strongly toward high redshift as expansion/activity stress increases and the growth coherence proxy decreases. Using Cosmic Chronometer data, we obtain a first observational projection that follows the same broad trend within the present uncertainties.

The result is not a new cosmological model. It is a compact diagnostic language for asking how expansion history and structure-growth history are structurally balanced across cosmic time. This suggests that cosmological evolution may be described not only as an expansion history, but as a structural-stress history that encodes the balance between dynamical growth and coherence formation.

References

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